

Inferential Statistics: Sampling & Confidence Intervals

1. What is Inferential Statistics?

Inferential statistics helps us **make conclusions or predictions about a population** using a **sample** of data.

- **Population** = everyone or everything we want to know about
 - **Sample** = a smaller group taken from the population
-

2. Sampling

Why Do We Sample?

Studying an entire population is often **impossible** (too large, expensive, time-consuming). A **good sample** can represent the whole population.

Types of Sampling

Type of Sampling	Description	Example
Random Sampling	Every member has equal chance	Pick 100 students randomly from a school
Stratified Sampling	Population divided into groups (age, gender), and sample is taken from each	50 boys and 50 girls
Systematic Sampling	Select every k-th member	Choose every 10th customer
Convenience Sampling	Easy to reach people (risky, biased)	Survey your classmates only

 **Important:** A good sample must be **random & unbiased** to correctly represent the population.

3. Sample Mean vs Population Mean

- **Population Mean (μ)** → True average of all data (unknown)
- **Sample Mean ($\bar{x}$)** → Average from our sample (known)

We use $\bar{x}$ to estimate μ .

4. Confidence Intervals (CI)

A **Confidence Interval** is a range of values that likely contains the true population mean.

Example: “We are 95% confident that the average height of students is between 160 cm and 170 cm.”



Basic Formula

Confidence Interval = $\bar{x} \pm \text{Margin of Error}$

Margin of Error depends on:

- Sample size (n)
- Standard deviation (σ or s)
- Confidence level (90%, 95%, 99%)



Common Confidence Levels

Confidence Level	Meaning
90%	Moderate certainty
95%	Most common in science
99%	Very high certainty
Higher confidence → Wider interval	
Lower confidence → Narrower interval	



Example (Simple)

A teacher takes a sample of **50 students**, average test score = **70**, margin of error = ± 5 .

95% Confidence Interval = $70 \pm 5 = [65, 75]$



Interpretation:

“We are 95% confident the true average score of ALL students is between **65 and 75.**”



Key Takeaways

Concept	Meaning
Sampling	Taking a small group to study a big population
Inferential Stats	Making predictions using sample data
Confidence Interval	Range that probably contains the true mean



Practical in Python — Sampling & Confidence Intervals



1. Setup Libraries

```
import numpy as np
import pandas as pd
from scipy import stats
import matplotlib.pyplot as plt
```



Explanation

- `numpy (np)` → Used for generating numerical data (populations, samples, calculations).
- `pandas (pd)` → Usually used for data tables (not heavily used here, but often part of analysis).
- `scipy.stats` → Provides statistical functions like confidence intervals.
- `matplotlib.pyplot` → Used to visualize data and confidence intervals.



Why Important?

These four libraries are the *core tools* for statistical analysis in Python.



2. Create a Population & Sample

```
infarential.ipynb - Day3 - Visual Studio Code
File Edit Selection View Go Run Terminal Help

Welcome | infarential.ipynb x
infarential.ipynb > ...
Generate + Code + Markdown | Run All Restart Clear All Outputs | Jupyter Variables ... Python 3.12.3

[1] ✓ 25.0s Python
import numpy as np
import pandas as pd
from scipy import stats
import matplotlib.pyplot as plt

[2] ✓ 0.0s Python
# Simulate population (heights in cm)
population = np.random.normal(170, 10, 10000)

# Take a random sample of 50
sample = np.random.choice(population, 50, replace=False)
print("Population Mean:", np.mean(population))
print("Sample Mean:", np.mean(sample))

... Population Mean: 169.99075274128398
Sample Mean: 168.7599935309455

Spaces: 4 Cell 5 of 5 Go Live {} 08:18
```

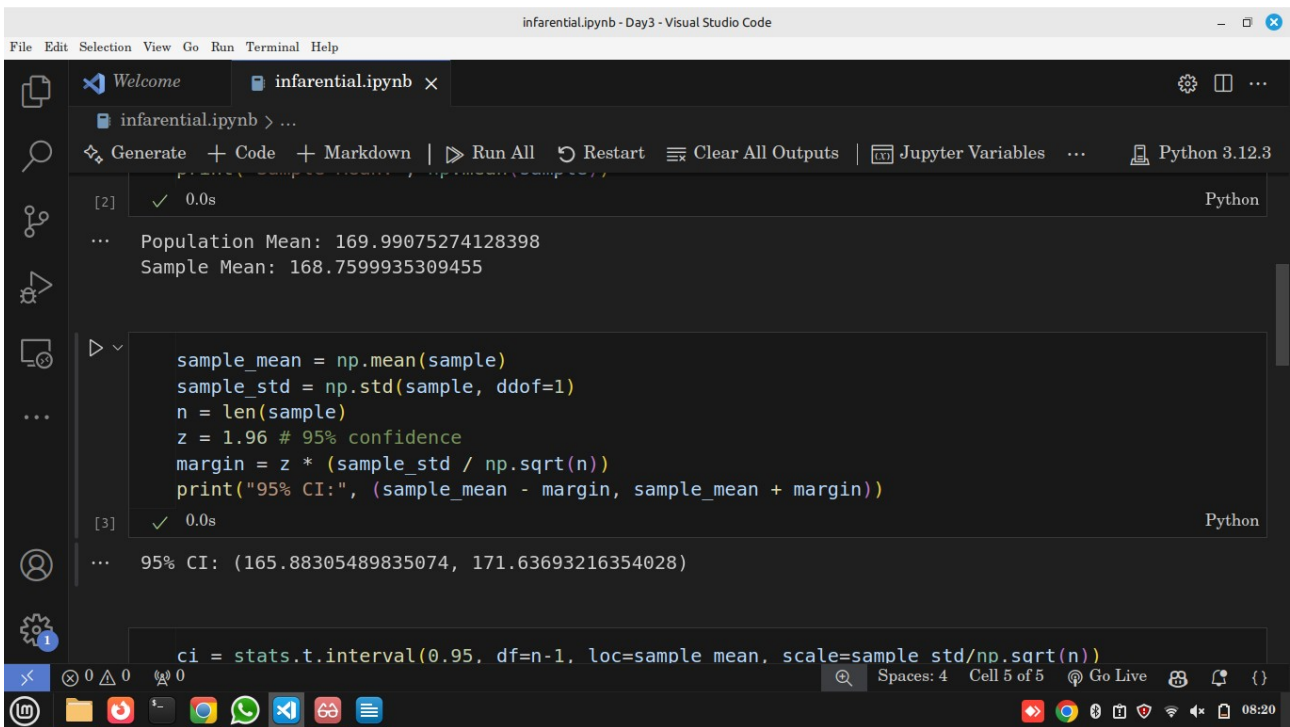
Explanation

- `np.random.normal(170, 10, 10000)`
→ Creates **10,000 people's heights**, mean = 170 cm, std = 10 cm. (Population).
- `np.random.choice(..., 50)`
→ Randomly selects **50 people** from the population (Sample).
- We print both Population Mean and Sample Mean to compare real vs estimated value.

Insight (Real World)

Just like surveys or opinion polls — we don't study everyone, we use a sample to estimate the population.

3. Manual Confidence Interval (95%)



The screenshot shows a Jupyter Notebook titled 'infarential.ipynb' in Visual Studio Code. The notebook contains two code cells. The first cell, labeled [2], calculates the population and sample means. The second cell, labeled [3], calculates the 95% confidence interval using the sample mean, standard deviation, and a Z-score of 1.96. The output of the second cell shows the 95% CI as (165.88305489835074, 171.63693216354028). A third line of code is visible at the bottom, using the `stats.t.interval` function to calculate the confidence interval.

```
infarential.ipynb > ...
[2] ✓ 0.0s Python
... Population Mean: 169.99075274128398
Sample Mean: 168.7599935309455

sample_mean = np.mean(sample)
sample_std = np.std(sample, ddof=1)
n = len(sample)
z = 1.96 # 95% confidence
margin = z * (sample_std / np.sqrt(n))
print("95% CI:", (sample_mean - margin, sample_mean + margin))

[3] ✓ 0.0s Python
... 95% CI: (165.88305489835074, 171.63693216354028)

ci = stats.t.interval(0.95, df=n-1, loc=sample mean, scale=sample std/np.sqrt(n))
```

Explanation

- `sample_mean` → Average height from our 50 people.
- `sample_std` → Variation within the sample (standard deviation).
- `n` → Sample size (50).
- `z = 1.96` → Z-score for **95% confidence**.
- `margin = z * (sample_std / sqrt(n))` → Formula for **margin of error**.
- Final CI = `sample_mean ± margin`.

💡 Insight

We are saying:

👉 “We are 95% confident the true average height of the population lies within this range.”



4. SciPy Confidence Interval (Shortcut)

The screenshot shows a Jupyter Notebook interface in Visual Studio Code. The notebook is titled 'infarential.ipynb'. The first cell (index 3) contains the output of a previous calculation: '95% CI: (165.88305489835074, 171.63693216354028)'. The second cell (index 4) contains the following Python code:

```
ci = stats.t.interval(0.95, df=n-1, loc=sample_mean, scale=sample_std/np.sqrt(n))
print("SciPy CI:", ci)
```

The output of the second cell is: 'SciPy CI: (165.81028708849257, 171.70969997339844)'. Below the code cell, there is a third cell with the following code:

```
plt.axvline(sample_mean, label='Sample Mean')
```

The bottom status bar of the VS Code window shows 'Spaces: 4', 'Cell 4 of 5', 'Go Live', and the time '08:21'.

📖 Explanation

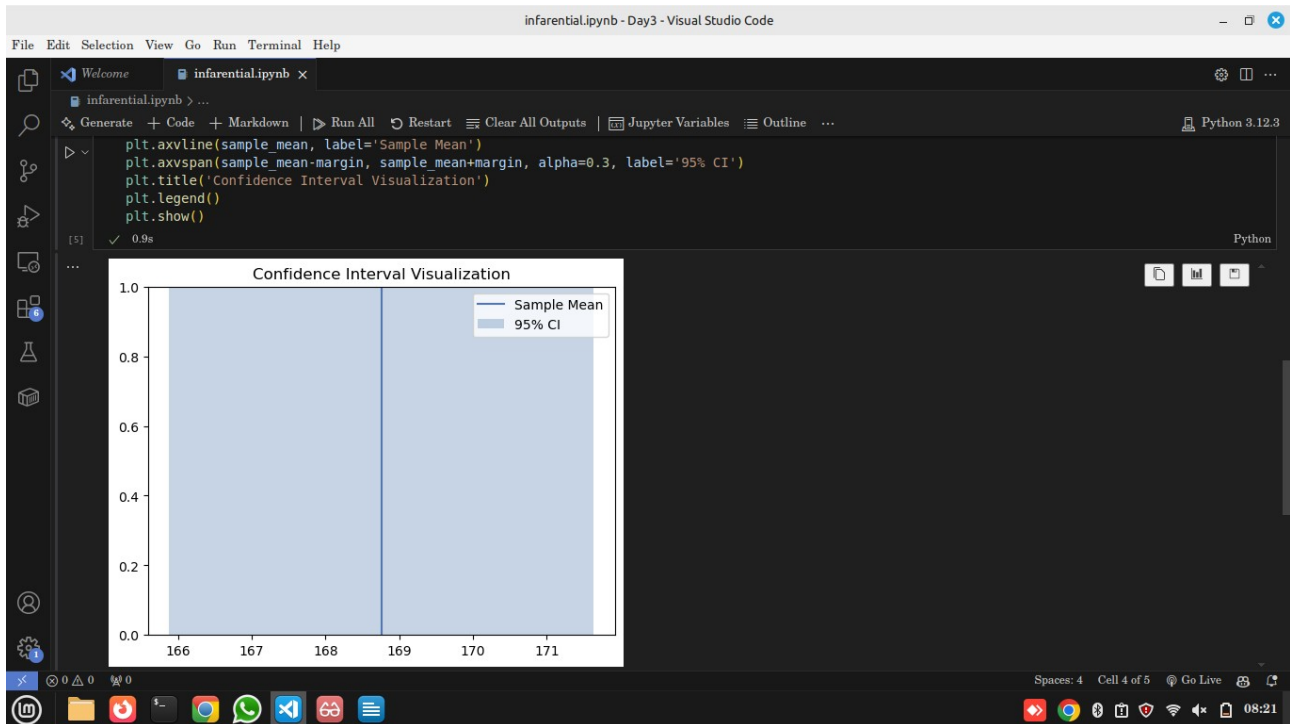
- `stats.t.interval` → Automatically calculates CI using the **t-distribution**.
- `0.95` → 95% confidence level.
- `df=n-1` → Degrees of freedom (49).
- `loc` → Sample mean.
- `scale` → Standard error ($\text{std} / \sqrt{n}$).

💡 Why Use SciPy?

You don't have to manually calculate — this is used in real data science projects for accuracy.



5. Visualize Confidence Interval



Explanation

- `axvline()` → Draws a vertical line at the sample mean.
- `axvspan()` → Shades the *confidence interval range*.
- `alpha=0.3` → Makes shading transparent.
- `legend()` → Shows labels.
- `show()` → Displays the plot.



Insight

The shaded area shows all possible values where the **true population mean** could lie with 95% confidence.

If we repeat sampling 100 times → about 95 intervals will contain the true mean.



Real-World Applications of Sampling & Confidence Intervals



1. Medical Research & Drug Testing

When developing a new medicine, it's impossible to test **every single patient on Earth**.

- ✓ Researchers test a **sample** of patients
- ✓ Calculate average recovery rate
- ✓ Use **confidence intervals** to estimate how effective the drug is in the entire population

💡 Example:

"We are 95% confident that this drug improves recovery time by 4 to 6 days."



2. Opinion Polls & Elections

News agencies don't interview every citizen before an election.

- ✓ They survey a **sample** of voters
- ✓ Calculate percentage support for a candidate
- ✓ Use confidence intervals to predict national results

💡 Example:

"Candidate A is expected to get $52\% \pm 3\%$ votes (95% CI $\rightarrow$ 49% to 55%)."



3. Customer Satisfaction & Market Research

Companies use surveys to test new products.

- ✓ They sample 1000 customers
- ✓ Estimate satisfaction level
- ✓ Use CI to infer how **millions** of customers might react

💡 Example:

"We are 95% confident that overall satisfaction is between 80% and 85%."



4. Manufacturing & Quality Control

Factories cannot test *every* product due to cost/time.

- ✓ They measure a sample of products (e.g. bottle weights)
- ✓ Use confidence intervals to estimate if machine is working correctly



Example:

"Bottle weight is 500 ml $\pm$ 2 ml (95% CI)."



5. Finance & Risk Estimation

Risk analysts compute average returns or losses using market samples.



Use sample data (historical prices)



Use CI to predict future earnings or risk range



Example:

"Expected portfolio return is 8% $\pm$ 1.5% (95% CI)."



Why Data Scientists Must Learn This

Concept

Real Use

Sampling

Saves time and cost, works with real-world limits

Confidence Interval

Measures uncertainty and reliability

Estimate Population Mean

Vital for prediction & decision making

Assignment



Time: 15 Minutes

- **Short Answer & Practical Tasks**
-



Part A – Conceptual (5 min)

Answer in 2–3 sentences each:

1. **Sampling Error:**
What is sampling error, and why does it occur in statistics?
 2. **Confidence Interval (CI):**
What does a 95% confidence interval really mean?
 3. **Sample vs Population:**
Why is it often impractical to collect data from the entire population?
-



Part B – Practical Application (7 min)

A company wants to know the **average monthly spending** of its customers. A sample of **100 customers** shows an average spending of **\$250** with a **standard deviation of \$40**.

1 Calculate the 95% Confidence Interval

Use formula:

$$\text{CI} = \text{Mean} \pm 1.96 \times (\text{SD} / \sqrt{n})$$

Show your steps.

- Mean = 250
- SD = 40
- n = 100

Questions:

- a) What is the margin of error?
- b) What is the confidence interval range?